RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

CHARGEPULSE: INDIA EV CHARGING DEMAND ANALYTICS

A Mini Project Report

Submitted by

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In partial fulfilment of the requirements for the course

Build and Deploy for Machine Learning

Academic project report · 2026

Abstract

This project develops ChargePulse, a compact offline analytics demonstration for estimating electric-vehicle charging sessions per station-hour. A transparent, deterministic synthetic dataset represents six illustrative city clusters and includes calendar, weather, event, and charger-capacity inputs. A NumPy least-squares linear regression is evaluated using a chronological 80/20 split and compared with a training-mean baseline. The project includes reproducible code, a local interactive interface, summary plots, and a downloadable dataset. Since the data are simulated, results demonstrate the workflow only and do not establish operational forecasting accuracy.

Keywords: EV charging, demand estimation, regression, reproducible analytics, synthetic data

1. Introduction

As electric mobility expands, charging operators need ways to reason about when local station capacity may be busy. This coursework project demonstrates a full analytics workflow: generate a structured dataset, define a prediction target, train a baseline model, evaluate on later observations, and expose a small local interface. The chosen theme differs from water-demand forecasting by focusing on station-hour charging sessions and capacity/weather context.

2. Problem statement and objectives

The task is to estimate the number of charging sessions in one station cluster during a specified hour using a small set of available context variables. Objectives are to create a consistent synthetic dataset, construct a time-aware evaluation, compare a regression model with a simple baseline, and make the result inspectable in a local application.

3. Data design and provenance

The supplied CSV contains 10,800 rows: 75 consecutive days × 24 hours × 6 illustrative station clusters. Each row represents one cluster-hour. Fields include timestamp, station cluster, approximate coordinates, hour, day of week, weekend flag, temperature, rainfall, port count, an illustrative event flag, session count, energy, and port utilization. The fixed random seed makes the generated file deterministic.

Important provenance statement: these observations are synthetic teaching data generated by the project; they are not measured charging-network data, public utility records, or a named third-party dataset. Weather and activity patterns are plausible inputs to an illustrative simulation and should not be interpreted as empirical city statistics.

Table 1. Dataset summary

Item	Description
Rows / period	10,800 station-hour records across 75 days
Clusters	Chennai, Bengaluru, Hyderabad, Mumbai, Delhi, Kochi illustrative clusters
Target	Charging sessions per station-hour
Inputs	Hour, weekend, temperature, rainfall, port count, event flag, latitude, longitude
Provenance	Deterministic synthetic data; not measured field data

4. Methodology

The rows are sorted by timestamp and split chronologically: the first 80% (8,640 rows) train the model and the final 20% (2,160 rows) form the holdout. Input columns are standardized using training-set means and standard deviations only. An intercept plus standardized predictors are fitted with NumPy least squares. The comparison baseline predicts the training target mean for every test row. Metrics are mean absolute error (MAE), root mean squared error (RMSE), and coefficient of determination (R^2).

This is a compact linear baseline, selected for transparency and dependency-light execution. It does not include hyperparameter search or claim optimized performance.

5. Results

On the final 2,160 held-out rows, the linear model achieved MAE 2.796 sessions, RMSE 3.498 sessions, and R^2 0.3523. The training-mean baseline MAE was 3.452 sessions. These figures are computed from the deterministic dataset bundled with this report.

Table 2. Held-out evaluation

Measure	Result	Interpretation
Test MAE	2.796 sessions	Average absolute prediction error on the chronological holdout
Test RMSE	3.498 sessions	Penalizes larger errors more strongly
Test R^2	0.3523	Relative fit on the holdout; may be negative if weaker than a mean predictor
Baseline MAE	3.452 sessions	Training-mean comparator

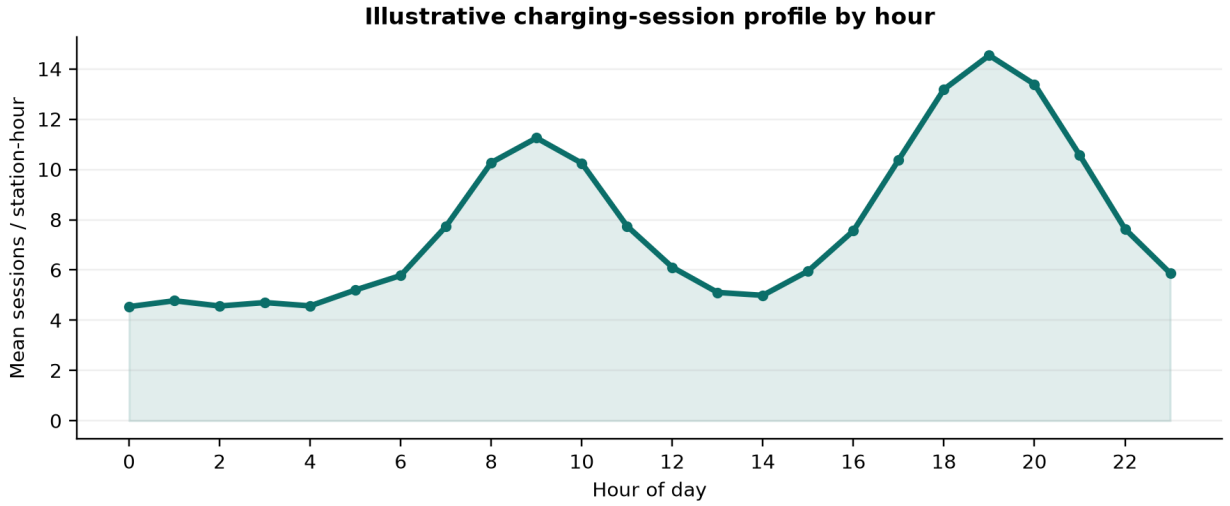


Figure 1. Hourly session profile averaged across the illustrative station clusters.

5. Results (continued)

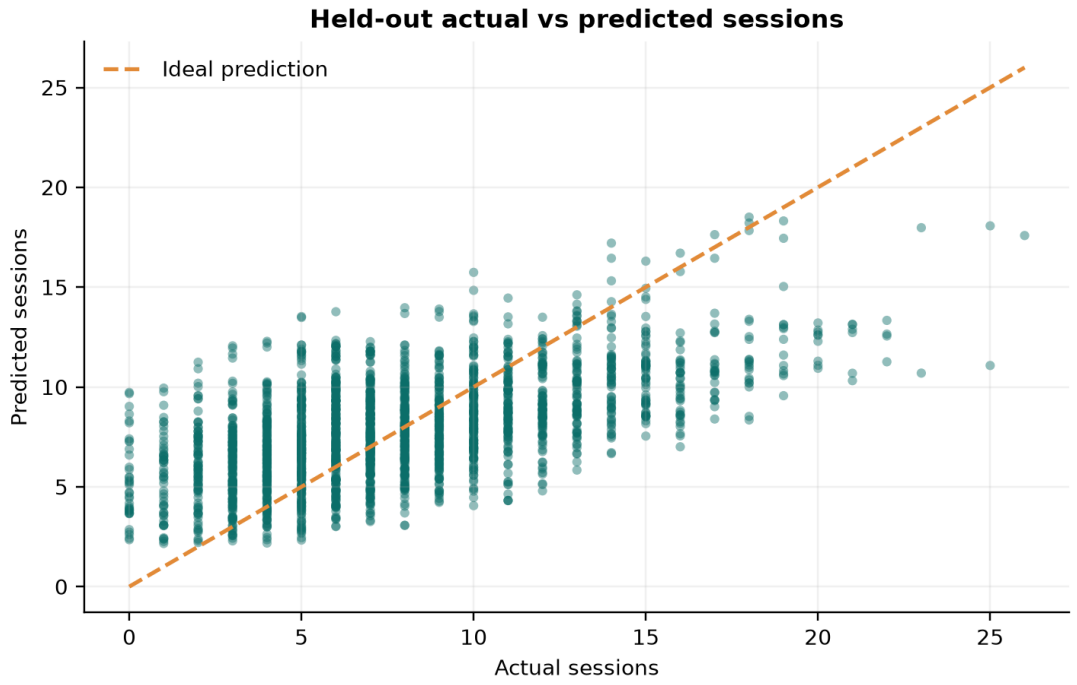


Figure 2. Model estimates versus actual simulated counts for the chronological test period; the dashed line denotes perfect agreement.

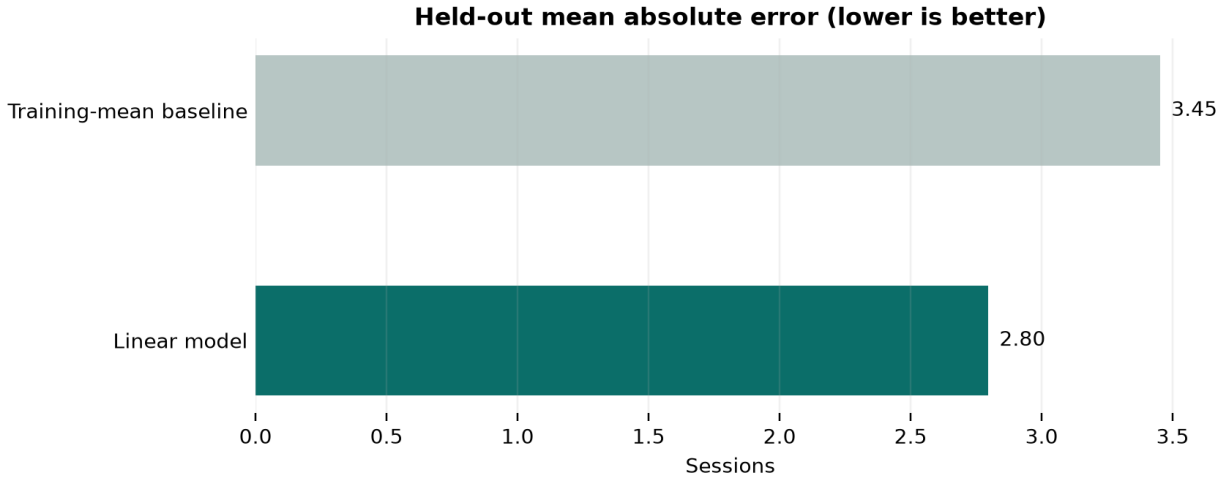


Figure 3. Mean absolute error comparison against the training-mean baseline. Lower is better.

6. Local application and deployment

The Streamlit interface (`mini_project/app.py`) accepts a cluster and user-selected hour, weekend, temperature, rainfall, and event context. It loads the CSV locally, fits the same NumPy model, and displays a session estimate with a clear simulated-data caveat. It does not call a remote service or need an API key. After Python dependencies have been installed, it can run offline on the same computer.

To reproduce: from the `mini_project` directory install `requirements.txt`; run `python run_analysis.py` to print the metrics; run `streamlit run app.py` to open the interactive local page. `metrics.json` stores the observed evaluation values. The charts in `assets/` are generated from the same data and model split.

7. Limitations and responsible interpretation

The data are simulated, small, and generated from a simplified process. City labels and coordinates identify illustrative clusters only. Event and weather inputs are synthetic. The fitted relationship is not validated against actual operators, charging transactions, or real meteorological records. The interface is a coursework prototype, not an operational planning tool. A real deployment would require permissioned charger-session data, documented data governance, leakage checks, stronger time-series validation, uncertainty intervals, monitoring, and stakeholder review.

8. Conclusion

ChargePulse presents an end-to-end, reproducible BDML exercise around EV charging demand: dataset construction, chronological evaluation, baseline comparison, visual analysis, and a local interactive demo. Its primary contribution is the transparent offline workflow and explicit data provenance. The results should be read as a software demonstration on synthetic observations, not as empirical claims about Indian charging demand.

References and resources

1. NumPy documentation, Linear algebra (`numpy.linalg.lstsq`): https://numpy.org/doc/stable/reference/generated/numpy.linalg.lstsq.html
2. Streamlit documentation, local app execution: https://docs.streamlit.io/
3. Dataset provenance: project-generated synthetic CSV and deterministic generator documented in `mini_project/README.md` and `mini_project/run_analysis.py`. No external dataset is claimed.

Appendix A. Project files

The GitHub repository contains the complete CSV, source code, requirements, metrics, and generated charts alongside this report. A clean checkout can reproduce the evaluation after installing the listed packages.